

# Chapter 5A:

## Algebra

8 lessons · one lesson for each exercise · each lesson is one hour

**NSW syllabus outcome**

Stage 4 Number and algebra: MA4-ALG-C-01 generalises number properties to operate with algebraic expressions.

CHAPTER

# 5A

**5.01** The commutative and associative laws

**5.02** The distributive law

**5.03** Variables and algebraic notation

**5.04** Expressions from words

**5.05** Substitution

**5.07** Adding and subtracting terms

**5.08** Multiplying terms

**5.09** Dividing terms

## How to use this booklet

**1** Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

**WE DO** **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

**DRILL** **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

**3** Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

**2** Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

**YOU DO** **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

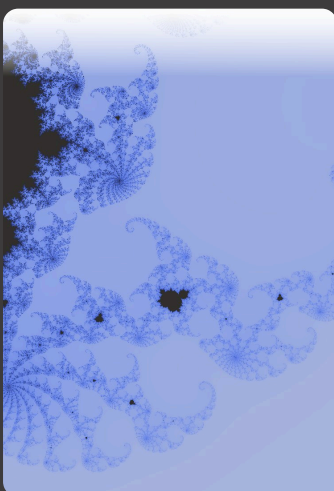
**4** Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

## Contents and progress tracker

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## In this chapter you will

- use the commutative, associative and distributive laws
- use variables and algebraic notation
- write expressions from words
- substitute into expressions and formulas
- add, subtract, multiply and divide algebraic terms



### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule,  $z \rightarrow z^2 + c$ , thousands of times for every point. Every booklet shows a different part of the set: this one is Elephant Valley, the row of trunks along the right-hand edge of the main body. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON  
**5.01**

# The commutative and associative laws

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- use the commutative and associative laws to make calculations easier

## Success criteria

I can:

- ☐ know that order does not matter for + and  $\times$  (commutative law)
- ☐ know that grouping does not matter for + and  $\times$  (associative law)
- ☐ regroup numbers to make friendly pairs such as 10s and 100s

**Key terms** Write the definition as your teacher explains it.

**Commutative**

**Associative**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Evaluate by pairing friendly numbers.

**a**  $7 + 18 + 3$

**b**  $91 + 18 + 9$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Evaluate by pairing friendly numbers.

14 QUESTIONS

**REMEMBER** Commutative:  $a + b = b + a$  and  $a \times b = b \times a$ .

**1**  $7 + 18 + 3$

**2**  $91 + 18 + 9$

**3**  $46 + 25 + 4$

**4**  $15 + 38 + 5$

**5**  $64 + 17 + 36$

**6**  $8 + 29 + 12 + 1$

Questions 7–14: pair friendly numbers.

**7**  $6 + 27 + 4$

**8**  $55 + 19 + 45$

**9**  $2 \times 43 \times 5$

**10**  $4 \times 9 \times 25$

**11**  $8 + 71 + 2$

**12**  $20 \times 7 \times 5$

**13**  $33 + 48 + 7$

**14**  $5 \times 13 \times 2$

## SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

## YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 36

Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–20: evaluate mentally. Pair the friendly numbers first.

1  $5 \times 26 \times 2$

\_\_\_\_\_

2  $41 \times 25 \times 4$

\_\_\_\_\_

3  $4 \times 34 \times 5$

\_\_\_\_\_

4  $3 \times 50 \times 2$

\_\_\_\_\_

5  $3 \times 5 \times 2$

\_\_\_\_\_

6  $2 \times 50 \times 28$

\_\_\_\_\_

7  $25 \times 4 \times 16$

\_\_\_\_\_

8  $4 \times 4 \times 5$

\_\_\_\_\_

9  $2 \times 26 \times 50$

\_\_\_\_\_

10  $29 \times 2 \times 50$

\_\_\_\_\_

11  $50 \times 2 \times 3$

\_\_\_\_\_

12  $125 \times 8 \times 41$

\_\_\_\_\_

13  $5 \times 22 \times 2$

\_\_\_\_\_

14  $20 \times 40 \times 5$

\_\_\_\_\_

15  $5 \times 2 \times 31$

\_\_\_\_\_

16  $2 \times 34 \times 50$

\_\_\_\_\_

17  $28 \times 2 \times 50$

\_\_\_\_\_

18  $4 \times 19 \times 5$

\_\_\_\_\_

19  $19 \times 125 \times 8$

\_\_\_\_\_

20  $47 \times 2 \times 50$

\_\_\_\_\_

## ROUND 2 Questions 21–36: true or false?

21  $104 \div 8 = 8 \div 104$

\_\_\_\_\_

22  $25 - 20 = 20 - 25$

\_\_\_\_\_

23  $42 \div 6 = 6 \div 42$

\_\_\_\_\_

24  $30 - 26 = 26 - 30$

\_\_\_\_\_

25  $30 - 14 = 14 - 30$

\_\_\_\_\_

26  $4 - 18 = 18 - 4$

\_\_\_\_\_

27  $14 - 12 = 12 - 14$

\_\_\_\_\_

28  $252 \div 21 = 21 \div 252$

\_\_\_\_\_

29  $780 \div 30 = 30 \div 780$

\_\_\_\_\_

30  $29 \times 28 = 28 \times 29$

\_\_\_\_\_

31  $27 \times 13 = 13 \times 27$

\_\_\_\_\_

32  $27 + 11 = 11 + 27$

\_\_\_\_\_

33  $68 \div 4 = 4 \div 68$

\_\_\_\_\_

34  $29 + 21 = 21 + 29$

\_\_\_\_\_

35  $56 \div 4 = 4 \div 56$

\_\_\_\_\_

36  $150 \div 25 = 25 \div 150$

\_\_\_\_\_

**YOU DO**

On your own: **Easy** Set B. Evaluate by pairing friendly numbers.

6 QUESTIONS

1  $2 \times 17 \times 5$

4  $4 \times 13 \times 25$

2  $25 \times 31 \times 4$

5  $50 \times 7 \times 2$

3  $5 \times 9 \times 20$

6  $2 \times 8 \times 5 \times 5$



**Level 2 · Building up**

Medium · 6 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

Evaluate by pairing friendly numbers.

a  $2 \times 17 \times 5$

Your teacher works this out. Copy every step.

b  $25 \times 31 \times 4$

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** Set C. Evaluate mentally.

6 QUESTIONS

**REMEMBER** Associative:  $(a + b) + c = a + (b + c)$ .

1  $13 \times 30$

3  $12 \times 50$

5  $600 \times 4$

2  $7 \times 400$

4  $30 \times 80$

6  $25 \times 12$

## SKILL DRILL 2 Medium basics: keep practising

32 questions

## YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 32 Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–20: evaluate mentally. Rearrange to make tens first.

1  $72 + 32 + 28$   
\_\_\_\_\_

2  $64 + 15 + 6$   
\_\_\_\_\_

3  $41 + 44 + 29$   
\_\_\_\_\_

4  $11 + 30 + 19$   
\_\_\_\_\_

5  $31 + 64 + 39$   
\_\_\_\_\_

6  $61 + 68 + 29$   
\_\_\_\_\_

7  $35 + 45 + 15$   
\_\_\_\_\_

8  $31 + 34 + 39$   
\_\_\_\_\_

9  $76 + 19 + 24$   
\_\_\_\_\_

10  $40 + 51 + 10$   
\_\_\_\_\_

11  $33 + 84 + 7$   
\_\_\_\_\_

12  $25 + 41 + 35$   
\_\_\_\_\_

13  $24 + 19 + 6$   
\_\_\_\_\_

14  $22 + 87 + 8$   
\_\_\_\_\_

15  $16 + 53 + 34$   
\_\_\_\_\_

16  $51 + 36 + 39$   
\_\_\_\_\_

17  $62 + 67 + 28$   
\_\_\_\_\_

18  $89 + 85 + 31$   
\_\_\_\_\_

19  $72 + 45 + 18$   
\_\_\_\_\_

20  $63 + 22 + 7$   
\_\_\_\_\_

## ROUND 2 Questions 21–32: which law is shown? Write commutative, associative or neither.

21  $(2 \times 6) \times 5 = 2 \times (6 \times 5)$   
\_\_\_\_\_

22  $a + b = b + a$   
\_\_\_\_\_

23  $(a + b) + c = a + (b + c)$   
\_\_\_\_\_

24  $8 \times 7 = 7 \times 8$   
\_\_\_\_\_

25  $ab = ba$   
\_\_\_\_\_

26  $9 + 14 = 14 + 9$   
\_\_\_\_\_

27  $(3 + 4) + 5 = 3 + (4 + 5)$   
\_\_\_\_\_

28  $12 - 5 \neq 5 - 12$   
\_\_\_\_\_

29  $15 - 6 - 2 \neq 15 - (6 - 2)$   
\_\_\_\_\_

30  $(xy)z = x(yz)$   
\_\_\_\_\_

31  $(20 \div 4) \div 5 \neq 20 \div (4 \div 5)$   
\_\_\_\_\_

32  $6 \times 3 \times 5 = 6 \times 5 \times 3$   
\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.**Example 3** Evaluate mentally.

**a**  $13 \times 30$

Your teacher works this out. Copy every step.

**b**  $7 \times 400$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

**REMEMBER** Look for pairs making 10, 100 or 1000.

- 1 Does the commutative law work for subtraction?  
Test it with  $10 - 4$ .

- 3 Evaluate  $125 \times 17 \times 8$  mentally.

- 2 Does the associative law work for division? Test it  
with  $(24 \div 6) \div 2$ .

- 4 Evaluate  $1 + 2 + 3 + \dots + 20$  by pairing numbers.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** Show off your skillChoose **ONE** question. Show  
your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**Evaluate  $16 + 27 + 4$ .**Level 2**Evaluate  $5 \times 23 \times 2$ .**Level 3**Evaluate  $20 \times 13 \times 5$ .

 Extension · Stretch yourself Worked example

Evaluate  $4 \times 7 \times 25 \times 3 \times 5 \times 2$  mentally.

1. Regroup:  $(4 \times 25) \times (5 \times 2) \times 7 \times 3$
2.  $100 \times 10 \times 21$

Answer: 21 000

**YOU DO**

On your own: read the worked example, then try these.

- E1** Evaluate  $8 \times 125 \times 9 \times 5 \times 2$  mentally.

- E2** Explain why  $38 + 47 + 62 + 53$  is easy to do mentally.

**Lesson summary**

Missed the lesson? Start here.

**Remember**

1. Commutative:  $a + b = b + a$  and  $a \times b = b \times a$ .
2. Associative:  $(a + b) + c = a + (b + c)$ .
3. Look for pairs making 10, 100 or 1000.
4. Not true for  $-$  and  $\div$ .

 Level 1

$$7 + 18 + 3$$

1.  $(7 + 3) + 18$

Answer: 28

 Level 2

$$25 \times 31 \times 4$$

1.  $(25 \times 4) \times 31$

Answer: 3100

 Level 3

$$13 \times 30$$

1.  $13 \times 3 \times 10$

Answer: 390

 Extension

$$4 \times 7 \times 25 \times 3 \times 5 \times 2$$

1.  $100 \times 10 \times 21$

Answer: 21 000

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LESSON  
5.02

## The distributive law

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- use the distributive law for mental arithmetic

## Success criteria

I can:

- ☐ split a number into tens and units, or a friendly number  $\pm$  a bit
- ☐ multiply each part, then add or subtract
- ☐ choose the easiest way to split

**Key terms** Write the definition as your teacher explains it.**Distributive law****Expand**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Use the distributive law.

**a**  $22 \times 9$  as  $22 \times (10 - 1)$

Your teacher works this out. Copy every step.

**b**  $17 \times 8$  as  $(10 + 7) \times 8$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Complete the working to evaluate.

14 QUESTIONS

**REMEMBER**  $a \times (b + c) = a \times b + a \times c$ .

**1**  $14 \times 6 = 10 \times 6 + 4 \times 6$

**2**  $23 \times 4 = 20 \times 4 + 3 \times 4$

**3**  $31 \times 7 = 30 \times 7 + 1 \times 7$

**4**  $19 \times 5 = 20 \times 5 - 1 \times 5$

**5**  $42 \times 3 = 40 \times 3 + 2 \times 3$

**6**  $29 \times 4 = 30 \times 4 - 1 \times 4$

Questions 7–14: use the distributive law.

**7**  $16 \times 5$

**8**  $21 \times 6$

**9**  $19 \times 4$

**10**  $32 \times 3$

**11**  $99 \times 7$

**12**  $41 \times 5$

**13**  $18 \times 9$

**14**  $102 \times 4$

## SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

**YOU DO** On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 36 Mark it with your teacher. Aim to beat your time next lesson.

**ROUND 1** Questions 1–20: evaluate by splitting the number:  $6 \times 23 = 6 \times 20 + 6 \times 3$ .

1  $3 \times 21$

\_\_\_\_\_

2  $5 \times 95$

\_\_\_\_\_

3  $3 \times 37$

\_\_\_\_\_

4  $5 \times 64$

\_\_\_\_\_

5  $4 \times 68$

\_\_\_\_\_

6  $9 \times 16$

\_\_\_\_\_

7  $6 \times 97$

\_\_\_\_\_

8  $3 \times 49$

\_\_\_\_\_

9  $6 \times 63$

\_\_\_\_\_

10  $5 \times 99$

\_\_\_\_\_

11  $9 \times 92$

\_\_\_\_\_

12  $9 \times 84$

\_\_\_\_\_

13  $9 \times 15$

\_\_\_\_\_

14  $9 \times 37$

\_\_\_\_\_

15  $5 \times 28$

\_\_\_\_\_

16  $8 \times 81$

\_\_\_\_\_

17  $4 \times 69$

\_\_\_\_\_

18  $4 \times 62$

\_\_\_\_\_

19  $4 \times 72$

\_\_\_\_\_

20  $8 \times 99$

\_\_\_\_\_

**ROUND 2** Questions 21–36: evaluate using a near ten:  $9 \times 34 = 10 \times 34 - 34$ .

21  $99 \times 43$

\_\_\_\_\_

22  $99 \times 45$

\_\_\_\_\_

23  $29 \times 50$

\_\_\_\_\_

24  $19 \times 39$

\_\_\_\_\_

25  $11 \times 54$

\_\_\_\_\_

26  $21 \times 13$

\_\_\_\_\_

27  $9 \times 13$

\_\_\_\_\_

28  $19 \times 14$

\_\_\_\_\_

29  $99 \times 24$

\_\_\_\_\_

30  $99 \times 20$

\_\_\_\_\_

31  $9 \times 60$

\_\_\_\_\_

32  $11 \times 38$

\_\_\_\_\_

33  $99 \times 49$

\_\_\_\_\_

34  $29 \times 51$

\_\_\_\_\_

35  $99 \times 35$

\_\_\_\_\_

36  $21 \times 37$

\_\_\_\_\_

**YOU DO**

On your own: **Easy** Set B. Use the distributive law to evaluate.

6 QUESTIONS

1  $22 \times 9$

4  $13 \times 12$

2  $17 \times 8$

5  $15 \times 7$

3  $27 \times 11$

6  $39 \times 3$



**Level 2 · Building up**

Medium · 6 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

Use the distributive law.

a  $27 \times 11$

Your teacher works this out. Copy every step.

b  $13 \times 12$

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** Set C. Use the distributive law. Show your working.

6 QUESTIONS

**REMEMBER** Split into tens and units:  $17 = 10 + 7$ .

1  $48 \times 6$

3  $101 \times 23$

5  $8 \times 199$

2  $99 \times 35$

4  $53 \times 12$

6  $65 \times 21$

**SKILL DRILL 2** Medium basics: keep practising

32 questions

**YOU DO**

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 32 Mark it with your teacher. Aim to beat your time next lesson.

**ROUND 1** Questions 1–16: fill in the missing number.

1  $4 \times 52 = 4 \times \square + 4 \times 2$

\_\_\_\_\_

2  $6 \times 51 = 6 \times \square + 6 \times 1$

\_\_\_\_\_

3  $4 \times 18 = 4 \times 20 - 4 \times \square$

\_\_\_\_\_

4  $5 \times 39 = 5 \times 40 - 5 \times \square$

\_\_\_\_\_

5  $9 \times 58 = 9 \times 60 - 9 \times \square$

\_\_\_\_\_

6  $4 \times 61 = 4 \times \square + 4 \times 1$

\_\_\_\_\_

7  $4 \times 53 = 4 \times \square + 4 \times 3$

\_\_\_\_\_

8  $6 \times 83 = 6 \times \square + 6 \times 3$

\_\_\_\_\_

9  $4 \times 58 = 4 \times 60 - 4 \times \square$

\_\_\_\_\_

10  $9 \times 31 = 9 \times \square + 9 \times 1$

\_\_\_\_\_

11  $7 \times 19 = 7 \times 20 - 7 \times \square$

\_\_\_\_\_

12  $6 \times 77 = 6 \times 80 - 6 \times \square$

\_\_\_\_\_

13  $9 \times 19 = 9 \times 20 - 9 \times \square$

\_\_\_\_\_

14  $7 \times 18 = 7 \times 20 - 7 \times \square$

\_\_\_\_\_

15  $5 \times 71 = 5 \times \square + 5 \times 1$

\_\_\_\_\_

16  $6 \times 81 = 6 \times \square + 6 \times 1$

\_\_\_\_\_

**ROUND 2** Questions 17–32: expand the brackets.

17  $4(n - 9)$

\_\_\_\_\_

18  $4(t - 4)$

\_\_\_\_\_

19  $5(c - 7)$

\_\_\_\_\_

20  $4(y - 3)$

\_\_\_\_\_

21  $4(k - 2)$

\_\_\_\_\_

22  $8(k - 11)$

\_\_\_\_\_

23  $2(c - 6)$

\_\_\_\_\_

24  $2(t + 6)$

\_\_\_\_\_

25  $7(b - 1)$

\_\_\_\_\_

26  $2(t - 6)$

\_\_\_\_\_

27  $7(c + 10)$

\_\_\_\_\_

28  $7(y + 7)$

\_\_\_\_\_

29  $4(m + 12)$

\_\_\_\_\_

30  $6(n + 4)$

\_\_\_\_\_

31  $5(a - 7)$

\_\_\_\_\_

32  $9(k - 4)$

\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Use the distributive law.

**a**  $48 \times 6$

Your teacher works this out. Copy every step.

**b**  $99 \times 35$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

**REMEMBER** Or near a friendly number:  $9 = 10 - 1$ .

**1** Evaluate  $7 \times 68 + 7 \times 32$  mentally.

**3** Show two different ways to evaluate  $18 \times 25$ .

**2** Evaluate  $15 \times 43 - 15 \times 33$  mentally.

**4** Explain why  $6 \times 97 = 6 \times 100 - 6 \times 3$ .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill** Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Evaluate  $12 \times 7$  using the distributive law.

**Level 2**

Evaluate  $19 \times 6$ .

**Level 3**

Evaluate  $102 \times 45$ .

 Extension · Stretch yourself Worked exampleEvaluate  $999 \times 47$  mentally.

1.  $999 = 1000 - 1$
2.  $47\ 000 - 47$

Answer: 46 953

## YOU DO

On your own: read the worked example, then try these.

- E1 Evaluate  $1001 \times 64$  mentally.

- E2 Evaluate  $37 \times 99 + 37$  mentally.

## Lesson summary

Missed the lesson? Start here.

## Remember

1.  $a \times (b + c) = a \times b + a \times c$ .
2. Split into tens and units:  $17 = 10 + 7$ .
3. Or near a friendly number:  $9 = 10 - 1$ .
4. Multiply each part, then add or subtract.

 Level 1

$22 \times 9$

1.  $22 \times 10 - 22$

Answer: 198

 Level 2

$17 \times 8$

1.  $80 + 56$

Answer: 136

 Level 3

$99 \times 35$

1.  $3500 - 35$

Answer: 3465

 Extension

$999 \times 47$

1.  $47\ 000 - 47$

Answer: 46 953

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LESSON  
5.03

# Variables and algebraic notation

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- use a variable to describe a pattern
- write and read algebraic notation

## Success criteria

I can:

- ☐ write a general rule using a variable
- ☐ simplify notation, e.g.  $5 \times a \times 2 = 10a$
- ☐ write a term in expanded form

**Key terms** Write the definition as your teacher explains it.

**Variable**

**Pronumeral**

**Term**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Write the rule, then simplify.

**a** Write the rule for:  $3 - (-3) = 2 \times 3$ ,  $5 - (-5) = 2 \times 5$ ,  
 $11 - (-11) = 2 \times 11$

Your teacher works this out. Copy every step.

**b** Simplify  $x + x + x + x$ .

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Simplify.

14 QUESTIONS

**REMEMBER** A **VARIABLE** is a letter that stands for a number.

1  $x + x + x + x$

2  $u \div 5$

3  $2y - y$

4  $5 \times a \times 2$

5  $m \times 7$

6  $p \times q$

Questions 7–14: simplify.

7  $a + a + a$

8  $4 \times b$

9  $y \times 3 \times 2$

10  $m \times n$

11  $k \times k$

12  $p \div 3$

13  $2 \times x \times y \times 5$

14  $t + t - t$

## SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

**YOU DO** On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 40 Mark it with your teacher. Aim to beat your time next lesson.

**ROUND 1** Questions 1–24: simplify. Write the number first, then the letters in alphabetical order.

1  $b \times 8$

\_\_\_\_\_

2  $a \times 11$

\_\_\_\_\_

3  $y \times p \times 7$

\_\_\_\_\_

4  $p \times 4$

\_\_\_\_\_

5  $t \times 6$

\_\_\_\_\_

6  $a \times a$

\_\_\_\_\_

7  $m \times m$

\_\_\_\_\_

8  $y \times k \times 12$

\_\_\_\_\_

9  $k \times 4$

\_\_\_\_\_

10  $11 \times a$

\_\_\_\_\_

11  $6 \times c$

\_\_\_\_\_

12  $9 \times a$

\_\_\_\_\_

13  $p \times p$

\_\_\_\_\_

14  $6 \times a$

\_\_\_\_\_

15  $k \times k$

\_\_\_\_\_

16  $m \times 3$

\_\_\_\_\_

17  $c \times c$

\_\_\_\_\_

18  $x \times k \times 9$

\_\_\_\_\_

19  $p \times m \times 8$

\_\_\_\_\_

20  $t \times t$

\_\_\_\_\_

21  $3 \times b$

\_\_\_\_\_

22  $t \times 12$

\_\_\_\_\_

23  $t \times a \times 6$

\_\_\_\_\_

24  $a \times 7$

\_\_\_\_\_

**ROUND 2** Questions 25–40: write in expanded form (with  $\times$  signs).

25  $10tx$

\_\_\_\_\_

26  $12c$

\_\_\_\_\_

27  $6mx$

\_\_\_\_\_

28  $7c$

\_\_\_\_\_

29  $11m$

\_\_\_\_\_

30  $3a$

\_\_\_\_\_

31  $9kp$

\_\_\_\_\_

32  $11k$

\_\_\_\_\_

33  $3m$

\_\_\_\_\_

34  $8kn$

\_\_\_\_\_

35  $3c$

\_\_\_\_\_

36  $6m$

\_\_\_\_\_

37  $5ct$

\_\_\_\_\_

38  $4p$

\_\_\_\_\_

39  $2ap$

\_\_\_\_\_

40  $9ab$

\_\_\_\_\_



**YOU DO** On your own: **Easy** Set B. Simplify.

6 QUESTIONS

1  $8 \times c \times b \times 3 \times a$

4  $3 \times y \times y$

2  $2 \times m \times w \times 10 \times w$

5  $a + a + b + b + b$

3  $k \times k \times k$

6  $4 \times d \div 2$

**Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Simplify.

a Simplify  $5 \times a \times 2$ .

Your teacher works this out. Copy every step.

b Simplify  $8 \times c \times b \times 3 \times a$ .

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Write in expanded form.

6 QUESTIONS

**REMEMBER** Leave out  $\times$ :  $3 \times a = 3a$ . Number first, letters in order.

1  $4mnk$

3  $9c^2$

5  $a^3b$

2  $\frac{6a}{b}$

4  $3xy^2$

6  $\frac{5m}{2n}$

## SKILL DRILL 2 Medium basics: keep practising

35 questions

## YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 35

Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–20: simplify.

1  $n \times 4 \times b \times b \times b$

\_\_\_\_\_

2  $n \times 4 \times k \times k \times k$

\_\_\_\_\_

3  $9 \times n \times n \times y$

\_\_\_\_\_

4  $m \div 7$

\_\_\_\_\_

5  $m \times p \times p$

\_\_\_\_\_

6  $c \div 4$

\_\_\_\_\_

7  $b \times n \times n$

\_\_\_\_\_

8  $b \times m \times m$

\_\_\_\_\_

9  $x \times 6 \times a \times a \times a$

\_\_\_\_\_

10  $k \times n \times n$

\_\_\_\_\_

11  $a \times x \times x$

\_\_\_\_\_

12  $c \div 5$

\_\_\_\_\_

13  $p \times 5 \times n \times n \times n$

\_\_\_\_\_

14  $y \times 5 \times a \times a \times a$

\_\_\_\_\_

15  $c \div 7$

\_\_\_\_\_

16  $c \div 8$

\_\_\_\_\_

17  $2 \times a \times a \times t$

\_\_\_\_\_

18  $8 \times k \times k \times n$

\_\_\_\_\_

19  $b \div 2$

\_\_\_\_\_

20  $b \times k \times k$

\_\_\_\_\_

## ROUND 2 Questions 21–35: write in expanded form.

21  $4b^2p$

\_\_\_\_\_

22  $6p^2t$

\_\_\_\_\_

23  $4a^2x$

\_\_\_\_\_

24  $3b^2m$

\_\_\_\_\_

25  $5k^3p$

\_\_\_\_\_

26  $3k^2m$

\_\_\_\_\_

27  $6k^3y$

\_\_\_\_\_

28  $7p^2t$

\_\_\_\_\_

29  $9a^2x$

\_\_\_\_\_

30  $3k^2n$

\_\_\_\_\_

31  $5m^3p$

\_\_\_\_\_

32  $5m^2t$

\_\_\_\_\_

33  $9b^3p$

\_\_\_\_\_

34  $7a^2t$

\_\_\_\_\_

35  $8n^3x$

\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Write in expanded form.

**a** Write  $4mnk$  in expanded form.

Your teacher works this out. Copy every step.

**b** Write  $9c^2$  and  $\frac{6a}{b}$  in expanded form.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Patterns and rules.

4 QUESTIONS

**REMEMBER**  $a \times a = a^2$ ;  $a \div 5 = a/5$ .

**1** Write a rule for:  $4 + 4 = 2 \times 4$ ,  $7 + 7 = 2 \times 7$ ,  $10 + 10 = 2 \times 10$

**3** Matchstick squares: 1 square uses 4 sticks, 2 squares use 7, 3 squares use 10. Find a rule for  $n$  squares.

**2** Write a rule for:  $3 \times 0 = 0$ ,  $8 \times 0 = 0$ ,  $15 \times 0 = 0$

**4** Is  $3a^2$  the same as  $(3a)^2$ ? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Simplify  $y + y + y$ .

**Level 2**

Simplify  $4 \times p \times 3 \times q$ .

**Level 3**

Write  $5a^2b$  in expanded form.

 Extension · Stretch yourself Worked example

Write a rule for the pattern:  $1 \times 3 = 2^2 - 1$ ,  $2 \times 4 = 3^2 - 1$ ,  $3 \times 5 = 4^2 - 1$

1. First number is  $n$ , second is  $n + 2$
2. The square is of  $n + 1$

Answer:  $n(n + 2) = (n + 1)^2 - 1$

## YOU DO

On your own: read the worked example, then try these.

- E1 Write a rule for:  $5 \times 5 - 4 \times 6 = 1$ ,  $8 \times 8 - 7 \times 9 = 1$ ,  $3 \times 3 - 2 \times 4 = 1$

- E2 Test your rule with a number of your own.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **variable** is a letter that stands for a number.
  2. Leave out  $\times$ :  $3 \times a = 3a$ . Number first, letters in order.
  3.  $a \times a = a^2$ ;  $a \div 5 = a/5$ .
  4. Expanded form shows every  $\times$ .

 Level 1

Rule for  $3 - (-3) = 2 \times 3 \dots$

1. Replace the number with  $a$

Answer:  $a - (-a) = 2a$

 Level 2

$5 \times a \times 2$

1.  $5 \times 2 = 10$

Answer:  $10a$

 Level 3

$2 \times m \times w \times 10 \times w$

1.  $20 \times m \times w^2$

Answer:  $20mw^2$

 Extension

$9c^2$

1.  $9 \times c \times c$

Answer:  $9 \times c \times c$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
5.04

## Expressions from words

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- write algebraic expressions from worded statements

## Success criteria

I can:

- ☐ know the words for  $+$ ,  $-$ ,  $\times$  and  $\div$  (sum, difference, product, quotient)
- ☐ write an expression using a variable
- ☐ use brackets when needed

**Key terms** Write the definition as your teacher explains it.**Expression****Consecutive**

Notes

## Level 1 · Getting started

Easy · 18 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Write an expression using  $N$ .**a** 3 times the number  $N$ 

Your teacher works this out. Copy every step.

**b** the difference between  $N$  and 5

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Write an expression. Use  $N$  for the number. 12 QUESTIONS**REMEMBER** Sum  $\rightarrow +$ , difference  $\rightarrow -$ .**1** the number plus 7

---

**2** 3 times the number

---

**3** the number minus 5

---

**4** the number divided by 4

---

**5** the next consecutive number

---

**6** one-third of the number

---

Questions 7–12: write an expression.

**7** 5 more than  $x$ 

---

**8**  $x$  less 8

---

**9** double  $x$ 

---

**10**  $x$  shared by 3

---

**11**  $x$  squared

---

**12** the product of  $x$  and  $y$ 

---

## SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

## YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 36

Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–21: write an expression.

1  $b$  decreased by 2

\_\_\_\_\_

2 double  $c$ 

\_\_\_\_\_

3  $c$  minus 3

\_\_\_\_\_

4  $c$  decreased by 4

\_\_\_\_\_

5 7 less than  $c$ 

\_\_\_\_\_

6  $c$  minus 2

\_\_\_\_\_

7 5 less than  $c$ 

\_\_\_\_\_

8  $a$  minus 3

\_\_\_\_\_

9  $a$  increased by 10

\_\_\_\_\_

10 half of  $a$ 

\_\_\_\_\_

11  $b$  divided by 7

\_\_\_\_\_

12  $a$  minus 4

\_\_\_\_\_

13  $n$  minus 11

\_\_\_\_\_

14 8 less than  $c$ 

\_\_\_\_\_

15 11 less than  $k$ 

\_\_\_\_\_

16  $a$  decreased by 12

\_\_\_\_\_

17  $a$  increased by 4

\_\_\_\_\_

18 half of  $n$ 

\_\_\_\_\_

19  $c$  squared

\_\_\_\_\_

20 12 less than  $b$ 

\_\_\_\_\_

21 6 less than  $t$ 

\_\_\_\_\_

ROUND 2 Questions 22–36: write an expression. Use  $n$  for "the number".

22 the number minus 10

\_\_\_\_\_

23 the number divided by 6

\_\_\_\_\_

24 9 more than the number

\_\_\_\_\_

25 the number minus 5

\_\_\_\_\_

26 7 more than the number

\_\_\_\_\_

27 the number plus 6

\_\_\_\_\_

28 the number minus 6

\_\_\_\_\_

29 the number divided by 7

\_\_\_\_\_

30 the number minus 3

\_\_\_\_\_

31 the number times itself

\_\_\_\_\_

32 the number divided by 9

\_\_\_\_\_

33 the number divided by 10

\_\_\_\_\_

34 11 more than the number

\_\_\_\_\_

35 2 more than the number

\_\_\_\_\_

36 the number minus 12

\_\_\_\_\_

**YOU DO** On your own: **Easy** Set B. Write an expression.

## 6 QUESTIONS

1 the sum of  $M$  and 3

4 the product of  $a$  and  $b$

2 5 more than  $B$

5 10 less than  $y$

3  $2H$  decreased by  $k$

⑥  $p$  shared equally among 6

 **Level 2 · Building up**

Medium · 6 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 2** Write an expression.

**a** the next consecutive number after  $N$

Your teacher works this out. Copy every step.

**b** one-third of  $N$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Medium** Set C. Write an expression.

## 6 QUESTIONS

**REMEMBER** Product  $\rightarrow \times$ , quotient  $\rightarrow \div$ .

① double the sum of  $x$  and 4

[illegible]

③ 7 less than 3 times  $w$

[illegible]

5 the number before  $n$

② the square of  $t$

| Circumstance                  | Justified (%) | Not justified (%) |
|-------------------------------|---------------|-------------------|
| If someone is attacking you   | 85            | 15                |
| If someone is threatening you | 75            | 25                |
| If someone is harassing you   | 65            | 35                |
| If someone is insulting you   | 55            | 45                |
| If someone is annoying you    | 15            | 85                |

④ half of the difference between  $m$  and 1

| Case No. | Case Name      | Case Address  | Case City     | Case State | Case Zip | Case Phone   | Case Email                 | Case Date  | Case Time | Case Status | Case Notes                                    |
|----------|----------------|---------------|---------------|------------|----------|--------------|----------------------------|------------|-----------|-------------|-----------------------------------------------|
| 1        | John Doe       | 123 Main St   | New York      | NY         | 10001    | 212-555-1234 | john.doe@example.com       | 2023-10-27 | 14:30     | Completed   | Initial assessment and data collection.       |
| 2        | Jane Smith     | 456 Elm St    | Los Angeles   | CA         | 90001    | 310-555-5678 | jane.smith@example.com     | 2023-10-28 | 10:00     | In Progress | Interview with subject and review of records. |
| 3        | Robert Johnson | 789 Oak St    | Chicago       | IL         | 60601    | 312-555-9012 | robert.johnson@example.com | 2023-10-29 | 09:00     | Pending     | Awaiting further information from the client. |
| 4        | Emily White    | 101 Pine St   | San Francisco | CA         | 94101    | 415-555-3456 | emily.white@example.com    | 2023-10-30 | 11:00     | Completed   | Final report and recommendations provided.    |
| 5        | Michael Brown  | 202 Cedar St  | Phoenix       | AZ         | 85001    | 602-555-7890 | michael.brown@example.com  | 2023-10-31 | 13:00     | In Progress | Conducting follow-up interviews and analysis. |
| 6        | Sarah Green    | 303 Birch St  | Philadelphia  | PA         | 19101    | 215-555-2345 | sarah.green@example.com    | 2023-11-01 | 10:30     | Pending     | Waiting for data from external sources.       |
| 7        | David Lee      | 404 Maple St  | San Diego     | CA         | 92101    | 619-555-6789 | david.lee@example.com      | 2023-11-02 | 15:00     | Completed   | Analysis complete, report drafted.            |
| 8        | Lisa King      | 505 Walnut St | Portland      | OR         | 97201    | 503-555-0123 | lisa.king@example.com      | 2023-11-03 | 09:30     | In Progress | Reviewing findings and preparing conclusions. |
| 9        | James Hall     | 606 Elm St    | Seattle       | WA         | 98101    | 206-555-4567 | james.hall@example.com     | 2023-11-04 | 12:00     | Pending     | Final review and approval process.            |
| 10       | Amanda Taylor  | 707 Oak St    | Denver        | CO         | 80201    | 303-555-8901 | amanda.taylor@example.com  | 2023-11-05 | 14:00     | Completed   | Project closed, all data archived.            |

⑥ the average of  $a$  and  $b$

| Case No. | Case Name      | Case Address | Case City     | Case State | Case Zip | Case Phone   | Case Email                 | Case Date  | Case Time | Case Status | Case Notes                                    |
|----------|----------------|--------------|---------------|------------|----------|--------------|----------------------------|------------|-----------|-------------|-----------------------------------------------|
| 1        | John Doe       | 123 Main St  | New York      | NY         | 10001    | 212-555-1234 | john.doe@example.com       | 2023-10-27 | 14:30     | Completed   | Initial assessment and data collection.       |
| 2        | Jane Smith     | 456 Elm St   | Los Angeles   | CA         | 90001    | 310-555-5678 | jane.smith@example.com     | 2023-10-28 | 10:00     | In Progress | Interview with subject and review of records. |
| 3        | Robert Johnson | 789 Oak St   | Chicago       | IL         | 60601    | 312-555-9012 | robert.johnson@example.com | 2023-10-29 | 09:00     | Pending     | Awaiting approval for field visit.            |
| 4        | Emily White    | 101 Pine St  | San Francisco | CA         | 94101    | 415-555-3456 | emily.white@example.com    | 2023-10-30 | 11:00     | Completed   | Final report and analysis completed.          |
| 5        | Michael Brown  | 202 Cedar St | Phoenix       | AZ         | 85001    | 602-555-7890 | michael.brown@example.com  | 2023-10-31 | 13:00     | In Progress | Conducting follow-up interviews.              |
| 6        | Sarah Green    | 303 Birch St | Philadelphia  | PA         | 19101    | 215-555-2345 | sarah.green@example.com    | 2023-11-01 | 15:00     | Pending     | Waiting for data from external source.        |
| 7        | David Lee      | 404 Maple St | San Diego     | CA         | 92101    | 619-555-6789 | david.lee@example.com      | 2023-11-02 | 10:30     | Completed   | Interview and data collection complete.       |
| 8        | Lisa King      | 505 Elm St   | Seattle       | WA         | 98101    | 206-555-0123 | lisa.king@example.com      | 2023-11-03 | 12:00     | In Progress | Reviewing collected data and reports.         |
| 9        | James Hall     | 606 Oak St   | Portland      | OR         | 97201    | 503-555-4567 | james.hall@example.com     | 2023-11-04 | 08:00     | Pending     | Approval pending for next phase.              |
| 10       | Amanda Taylor  | 707 Pine St  | Denver        | CO         | 80201    | 303-555-8901 | amanda.taylor@example.com  | 2023-11-05 | 16:00     | Completed   | Final review and report submission.           |

SKILL DRILL 2 Medium basics: keep practising

27 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 27

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: write an expression.

1 the product of  $c$  and  $p$ , plus 2

\_\_\_\_\_

2 the sum of  $k$  and  $x$ , divided by 9

\_\_\_\_\_

3 4 less than twice  $k$

\_\_\_\_\_

4 twice the sum of  $a$  and 8

\_\_\_\_\_

5 the product of  $b$  and  $k$ , plus 3

\_\_\_\_\_

6 the sum of  $k$  and  $x$ , divided by 10

\_\_\_\_\_

7  $a$  squared, plus 11

\_\_\_\_\_

8 5 times  $a$ , take away 3

\_\_\_\_\_

9 7 times  $a$ , take away 7

\_\_\_\_\_

10 3 times  $b$ , take away 9

\_\_\_\_\_

11 5 less than the product of  $a$  and  $b$

\_\_\_\_\_

12 7 less than twice  $b$

\_\_\_\_\_

13 4 less than the product of  $p$  and  $t$

\_\_\_\_\_

14 3 less than the product of  $c$  and  $k$

\_\_\_\_\_

15 7 more than 4 times  $a$

\_\_\_\_\_

ROUND 2 Questions 16–27: write an expression for each amount.

16 the cost of  $y$  pens at \$13 each

\_\_\_\_\_

17 \$70 shared equally by  $k$  people

\_\_\_\_\_

18 the number of legs on  $k$  spiders

\_\_\_\_\_

19  $y$  minutes plus 10 minutes

\_\_\_\_\_

20  $b$  minutes plus 2 minutes

\_\_\_\_\_

21 the cost of  $b$  pens at \$11 each

\_\_\_\_\_

22  $b$  minutes plus 9 minutes

\_\_\_\_\_

23  $p$  minutes plus 2 minutes

\_\_\_\_\_

24 \$120 shared equally by  $t$  people

\_\_\_\_\_

25 the cost of  $p$  pens at \$8 each

\_\_\_\_\_

26 \$100 shared equally by  $t$  people

\_\_\_\_\_

27 the cost of  $m$  pens at \$6 each

\_\_\_\_\_



**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Write an expression.

**a**  $2H$  decreased by  $k$

Your teacher works this out. Copy every step.

**b** double the sum of  $x$  and 4

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Write an expression for each situation.

4 QUESTIONS

**REMEMBER** More than  $\rightarrow +$ , less than  $\rightarrow -$  (order!).

**1** The cost of  $n$  pens at \$3 each.

**3** A rectangle is  $L$  cm long and 5 cm wide. Write its perimeter.

**2** Sam is  $s$  years old. His sister is 4 years older. Find the sum of their ages.

**4** The change from \$50 after buying  $k$  drinks at \$2.50 each.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Write “the number plus 12” using  $N$ .

**Level 2**

Write “4 less than  $y$ ”.

**Level 3**

Write “the product of  $m$  and 5, then add 2”.

 Extension · Stretch yourself Worked example

Write an expression for the sum of three consecutive numbers starting at  $n$ , and simplify it.

1.  $n + (n + 1) + (n + 2)$
2. Collect like terms

Answer:  $3n + 3$

## YOU DO

On your own: read the worked example, then try these.

- E1 Write and simplify the sum of three consecutive even numbers starting at  $e$ .

- E2 Explain why the sum of 3 consecutive numbers is always divisible by 3.

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Sum  $\rightarrow +$ , difference  $\rightarrow -$ .
  3. More than  $\rightarrow +$ , less than  $\rightarrow -$  (order!).

2. Product  $\rightarrow \times$ , quotient  $\rightarrow \div$ .
4. Use brackets when a whole group is multiplied.

 Level 1

3 times  $N$

1.  $3 \times N$

Answer:  $3N$

 Level 2

$N$  decreased by 5

1.  $N - 5$

Answer:  $N - 5$

 Level 3

Double the sum of  $x$  and 4

1. Brackets first

Answer:  $2(x + 4)$

 Extension

3 consecutive numbers

1.  $n + n + 1 + n + 2$

Answer:  $3n + 3$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON  
5.05

# Substitution

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- find the value of an expression by substituting numbers

## Success criteria

I can:

- ☐ replace each variable with its value, using brackets
- ☐ follow the order of operations
- ☐ substitute into a formula

**Key terms** Write the definition as your teacher explains it.

**Substitute**

**Evaluate**

**Formula**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.

**Example 1** Substitute  $a = 2$ ,  $b = 5$  and  $c = 6$ .

**a** If  $a = 2$  and  $b = 5$ , find  $a + b$ .

**b** If  $a = 2$  and  $c = 6$ , find  $3a - c$ .

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: If  $a = 2$ ,  $b = 5$  and  $c = 6$ , evaluate:

14 QUESTIONS

**REMEMBER** Write the expression.

1  $a + b$

2  $c - a$

3  $4b$

4  $bc$

5  $a + b + c$

6  $3a - c$

Questions 7–14: if  $x = 3$  and  $y = 4$ , evaluate.

7  $x + y$

8  $2x$

9  $xy$

10  $y - x$

11  $5y$

12  $x^2$

13  $3x + y$

14  $10 - y$

## SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

## YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 36

Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–24: substitute, then evaluate.

1  $p + 3$ , when  $p = 2$

\_\_\_\_\_

2  $9x + 5$ , when  $x = 5$

\_\_\_\_\_

3  $5n$ , when  $n = 8$

\_\_\_\_\_

4  $8p$ , when  $p = 3$

\_\_\_\_\_

5  $8x$ , when  $x = 10$

\_\_\_\_\_

6  $5a$ , when  $a = 9$

\_\_\_\_\_

7  $p + 9$ , when  $p = 5$

\_\_\_\_\_

8  $p - 7$ , when  $p = 5$

\_\_\_\_\_

9  $3p$ , when  $p = 9$

\_\_\_\_\_

10  $4p + 1$ , when  $p = 12$

\_\_\_\_\_

11  $7p + 3$ , when  $p = 6$

\_\_\_\_\_

12  $n + 4$ , when  $n = 12$

\_\_\_\_\_

13  $4x$ , when  $x = 5$

\_\_\_\_\_

14  $n + 2$ , when  $n = 12$

\_\_\_\_\_

15  $6x$ , when  $x = 4$

\_\_\_\_\_

16  $8a + 4$ , when  $a = 8$

\_\_\_\_\_

17  $3n$ , when  $n = 3$

\_\_\_\_\_

18  $x + 2$ , when  $x = 10$

\_\_\_\_\_

19  $p + 6$ , when  $p = 11$

\_\_\_\_\_

20  $5a$ , when  $a = 6$

\_\_\_\_\_

21  $5n$ , when  $n = 3$

\_\_\_\_\_

22  $p + 4$ , when  $p = 8$

\_\_\_\_\_

23  $x + 2$ , when  $x = 9$

\_\_\_\_\_

24  $a + 6$ , when  $a = 5$

\_\_\_\_\_

ROUND 2 Questions 25–36: if  $a = 4$  and  $b = 7$ , evaluate.

25  $10 - a$

\_\_\_\_\_

26  $4b - 3a$

\_\_\_\_\_

27  $3b$

\_\_\_\_\_

28  $b - a$

\_\_\_\_\_

29  $2ab$

\_\_\_\_\_

30  $ab - 10$

\_\_\_\_\_

31  $b^2$

\_\_\_\_\_

32  $a + b$

\_\_\_\_\_

33  $a + b + 9$

\_\_\_\_\_

34  $2a + b$

\_\_\_\_\_

35  $ab$

\_\_\_\_\_

36  $5a - b$

\_\_\_\_\_

**YOU DO**

**On your own:** **Easy** Set B. If  $a = 2$ ,  $b = 5$  and  $c = 6$ , evaluate:

## 6 QUESTIONS

①  $a + 2b - c$

④  $\frac{c}{a}$

2 *abc*

5  $2(a + b)$

③  $c^2$

⑥  $b^2 - c$

 **Level 2 · Building up**

Medium · 6 questions

**WE DO**

**With your teacher: copy what they write on the board.**

### Example 2

**Substitute, then use the formula  $C = 30h + 19$ .**

**a** If  $a = 2$ ,  $b = 5$ ,  $c = 6$ , find  $abc$ .

Your teacher works this out. Copy every step.

**b** If  $C = 30h + 19$ , find  $C$  when  $h = 3$ .

Your teacher works this out. Copy every step.

**YOU DO**

**On your own:** **Medium** Set C. If  $C = 30h + 19$ , find C when:

## 6 QUESTIONS

**REMEMBER** Replace each variable with its value in brackets.

①  $h = 3$

| Patient Information   |  |
|-----------------------|--|
| First Name            |  |
| Last Name             |  |
| Address               |  |
| City                  |  |
| State                 |  |
| Zip                   |  |
| Phone                 |  |
| Insurance             |  |
| Physician Information |  |
| Physician Name        |  |
| Physician Address     |  |
| Physician City        |  |
| Physician State       |  |
| Physician Zip         |  |
| Physician Phone       |  |
| Physician Insurance   |  |
| Referral Information  |  |
| Referral Number       |  |
| Referral Date         |  |
| Referral Type         |  |
| Referral Source       |  |
| Referral Reason       |  |
| Referral Status       |  |
| Referral Notes        |  |
| Referral History      |  |
| Referral Date         |  |
| Referral Type         |  |
| Referral Source       |  |
| Referral Reason       |  |
| Referral Status       |  |
| Referral Notes        |  |
| Referral Summary      |  |
| Referral Date         |  |
| Referral Type         |  |
| Referral Source       |  |
| Referral Reason       |  |
| Referral Status       |  |
| Referral Notes        |  |

③  $h = 5$

⑤  $h = 0$

②  $h = -1$

④  $h = 2.5$

| Case No. | Case Name      | Case Address | Case City     | Case State | Case Zip | Case Phone   | Case Email                 | Case Date  | Case Time | Case Status | Case Notes                                    |
|----------|----------------|--------------|---------------|------------|----------|--------------|----------------------------|------------|-----------|-------------|-----------------------------------------------|
| 1        | John Doe       | 123 Main St  | New York      | NY         | 10001    | 212-555-1234 | john.doe@example.com       | 2023-10-27 | 14:30     | Completed   | Initial assessment and data collection.       |
| 2        | Jane Smith     | 456 Elm St   | Los Angeles   | CA         | 90001    | 310-555-5678 | jane.smith@example.com     | 2023-10-28 | 10:00     | In Progress | Interview with subject and review of records. |
| 3        | Robert Johnson | 789 Oak St   | Chicago       | IL         | 60601    | 312-555-9012 | robert.johnson@example.com | 2023-10-29 | 09:00     | Pending     | Awaiting approval for field visit.            |
| 4        | Maria Garcia   | 101 Pine St  | San Antonio   | TX         | 78201    | 210-555-3456 | maria.garcia@example.com   | 2023-10-30 | 11:00     | Completed   | Final report and analysis completed.          |
| 5        | David Lee      | 202 Birch St | Phoenix       | AZ         | 85001    | 602-555-7890 | david.lee@example.com      | 2023-10-31 | 13:00     | In Progress | Conducting follow-up interviews.              |
| 6        | Sarah Brown    | 303 Cedar St | San Diego     | CA         | 92101    | 619-555-2345 | sarah.brown@example.com    | 2023-11-01 | 15:00     | Pending     | Waiting for data from other sources.          |
| 7        | Michael White  | 404 Maple St | San Jose      | CA         | 95101    | 408-555-6789 | michael.white@example.com  | 2023-11-02 | 10:30     | Completed   | Interview and data collection complete.       |
| 8        | Emily Green    | 505 Elm St   | San Francisco | CA         | 94101    | 415-555-0123 | emily.green@example.com    | 2023-11-03 | 12:00     | In Progress | Reviewing collected data and reports.         |
| 9        | James Black    | 606 Oak St   | San Jose      | CA         | 95101    | 408-555-4567 | james.black@example.com    | 2023-11-04 | 14:00     | Pending     | Awaiting final review and approval.           |
| 10       | Alice Johnson  | 707 Pine St  | San Jose      | CA         | 95101    | 408-555-8901 | alice.johnson@example.com  | 2023-11-05 | 16:00     | Completed   | Final report and analysis completed.          |

⑥  $h = 10$

## SKILL DRILL 2 Medium basics: keep practising

28 questions

## YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 28 Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–16: substitute, then evaluate. Take care with negatives.

①  $x^2 + x, x = 3$

\_\_\_\_\_

②  $10 - x, x = -1$

\_\_\_\_\_

③  $4 - 2x, x = -4$

\_\_\_\_\_

④  $10 - x, x = 2$

\_\_\_\_\_

⑤  $10 - x, x = 1$

\_\_\_\_\_

⑥  $3x, x = 2$

\_\_\_\_\_

⑦  $x^2 + x, x = -6$

\_\_\_\_\_

⑧  $4 - 2x, x = 1$

\_\_\_\_\_

⑨  $5x + 1, x = -5$

\_\_\_\_\_

⑩  $x^2, x = 2$

\_\_\_\_\_

⑪  $3x, x = 3$

\_\_\_\_\_

⑫  $x + 8, x = -6$

\_\_\_\_\_

⑬  $5x + 1, x = -2$

\_\_\_\_\_

⑭  $x^2 + x, x = -2$

\_\_\_\_\_

⑮  $x^2, x = 5$

\_\_\_\_\_

⑯  $2x - 5, x = 5$

\_\_\_\_\_

## ROUND 2 Questions 17–28: substitute into the formula.

⑰  $s = d \div t, d = 108, t = 9$

\_\_\_\_\_

⑱  $A = lw, l = 11, w = 11$

\_\_\_\_\_

⑲  $C = 30h + 19, h = 6$

\_\_\_\_\_

⑳  $P = 2l + 2w, l = 14, w = 2$

\_\_\_\_\_

㉑  $P = 2l + 2w, l = 15, w = 3$

\_\_\_\_\_

㉒  $s = d \div t, d = 90, t = 9$

\_\_\_\_\_

㉓  $s = d \div t, d = 108, t = 12$

\_\_\_\_\_

㉔  $A = lw, l = 4, w = 5$

\_\_\_\_\_

㉕  $s = d \div t, d = 24, t = 3$

\_\_\_\_\_

㉖  $P = 2l + 2w, l = 7, w = 9$

\_\_\_\_\_

㉗  $A = lw, l = 5, w = 11$

\_\_\_\_\_

㉘  $s = d \div t, d = 90, t = 10$

\_\_\_\_\_

Level 3 · Challenge

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Substitute into  $C = 30h + 19$ .

**a** Find  $C$  when  $h = -1$ .

Your teacher works this out. Copy every step.

**b** Find  $C$  when  $h = 2.5$ .

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Substitute into each formula.

4 QUESTIONS

**REMEMBER** Use the order of operations.

**1**  $P = 2L + 2W$ . Find  $P$  when  $L = 8$  and  $W = 3$ .

**3**  $F = 1.8C + 32$ . Find  $F$  when  $C = 25$ .

**2**  $A = \frac{1}{2}bh$ . Find  $A$  when  $b = 10$  and  $h = 7$ .

**4** If  $x = -3$ , evaluate  $x^2 + 2x$ .

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** Show off your skill

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

If  $m = 4$ , evaluate  $5m$ .

**Level 2**

If  $x = 3$  and  $y = 7$ , evaluate  $2x + y$ .

**Level 3**

If  $A = l^2 - 3l$ , find  $A$  when  $l = -2$ .

 Extension · Stretch yourself Worked example

A taxi charges  $C = 4 + 2.5k$  dollars for  $k$  km. How far can you go for \$39?

1.  $39 = 4 + 2.5k$
2.  $35 = 2.5k$
3.  $k = 35 \div 2.5$

Answer: 14 km

**YOU DO**

On your own: read the worked example, then try these.

- E1** Using the same formula, find the cost of a 22 km trip.

- E2** A phone plan costs  $C = 15 + 0.2m$  for  $m$  minutes. How many minutes for \$27?

**Lesson summary**

Missed the lesson? Start here.

**Remember**

1. Write the expression.
2. Replace each variable with its value in brackets.
3. Use the order of operations.
4. Remember  $3a$  means  $3 \times a$ .

 Level 1

$$a + b, a = 2, b = 5$$

1.  $2 + 5$

Answer: 7

 Level 2

$$3a - c$$

1.  $3 \times 2 - 6$

Answer: 0

 Level 3

$$C = 30h + 19, h = -1$$

1.  $30 \times (-1) + 19$

Answer: -11

 Extension

Taxi for \$39

1.  $2.5k = 35$

Answer: 14 km

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LESSON  
5.07

## Adding and subtracting terms

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- simplify expressions by collecting like terms

## Success criteria

I can:

- ☐ identify **like terms** (same letters, same powers)
- ☐ add and subtract the coefficients of like terms
- ☐ collect several sets of like terms

**Key terms** Write the definition as your teacher explains it.**Like terms****Coefficient**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Simplify by collecting like terms.

**a**  $3m + 4m$

Your teacher works this out. Copy every step.

**b**  $12m - 4m$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Simplify.

14 QUESTIONS

**REMEMBER** **LIKE TERMS** have exactly the same letters and powers.

**1**  $3m + 4m$

**2**  $12m - 4m$

**3**  $5a + a$

**4**  $9k - 9k$

**5**  $2x + 3x + 4x$

**6**  $7y - 4y - y$

Questions 7–14: simplify.

**7**  $4a + 3a$

**8**  $9x - 5x$

**9**  $2m + m$

**10**  $6k - k$

**11**  $3p + 2q + p$

**12**  $8y - 3y + y$

**13**  $5ab + 2ab$

**14**  $7c - 7c$

## SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

## YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 40

Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–24: simplify by adding or subtracting like terms.

1  $10n - 5n$   
\_\_\_\_\_

2  $6p - 8p$   
\_\_\_\_\_

3  $3t + t$   
\_\_\_\_\_

4  $c + 12c$   
\_\_\_\_\_

5  $4y - 2y$   
\_\_\_\_\_

6  $k + 11k$   
\_\_\_\_\_

7  $3m + 6m$   
\_\_\_\_\_

8  $12p - 6p$   
\_\_\_\_\_

9  $12t + 9t$   
\_\_\_\_\_

10  $12x - 7x$   
\_\_\_\_\_

11  $2m - 3m$   
\_\_\_\_\_

12  $6y - 2y$   
\_\_\_\_\_

13  $5c - 7c$   
\_\_\_\_\_

14  $7a - a$   
\_\_\_\_\_

15  $5p - 10p$   
\_\_\_\_\_

16  $7x - 11x$   
\_\_\_\_\_

17  $3c - c$   
\_\_\_\_\_

18  $8n + 9n$   
\_\_\_\_\_

19  $11n - 10n$   
\_\_\_\_\_

20  $4n + 4n$   
\_\_\_\_\_

21  $8n + 5n$   
\_\_\_\_\_

22  $7k + 8k$   
\_\_\_\_\_

23  $7x - 10x$   
\_\_\_\_\_

24  $a - 8a$   
\_\_\_\_\_

## ROUND 2 Questions 25–40: like terms? Write yes or no.

25  $nt$  and  $9tn$   
\_\_\_\_\_

26  $tx$  and  $4xt$   
\_\_\_\_\_

27  $bc$  and  $9cb$   
\_\_\_\_\_

28  $2n$  and  $9n$   
\_\_\_\_\_

29  $bx$  and  $9xb$   
\_\_\_\_\_

30  $7a$  and  $3n$   
\_\_\_\_\_

31  $7p$  and  $3x$   
\_\_\_\_\_

32  $c^2$  and  $8c$   
\_\_\_\_\_

33  $5m$  and  $8m$   
\_\_\_\_\_

34  $kn$  and  $6nk$   
\_\_\_\_\_

35  $8t$  and  $8t$   
\_\_\_\_\_

36  $5m$  and  $9m$   
\_\_\_\_\_

37  $p^2$  and  $8p$   
\_\_\_\_\_

38  $ap$  and  $3pa$   
\_\_\_\_\_

39  $cy$  and  $8yc$   
\_\_\_\_\_

40  $9a$  and  $4x$   
\_\_\_\_\_

**YOU DO**

On your own: **Easy** Set B. Are these like terms? Write yes or no.

6 QUESTIONS

1  $3a$  and  $5a$

4  $m^2$  and  $3m$

2  $2x$  and  $2y$

5  $6$  and  $6p$

3  $4ab$  and  $7ba$

6  $-2k^2$  and  $k^2$



**Level 2 · Building up**

Medium · 6 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

Simplify by collecting like terms.

a  $7y - 4y - y$

Your teacher works this out. Copy every step.

b  $6uv + 4 + 12uv$

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** Set C. Simplify.

6 QUESTIONS

**REMEMBER** Add or subtract only like terms.

1  $6uv + 4 + 12uv$

3  $8x - 3 + 2x + 7$

5  $10 - 3m - 4 + 5m$

2  $5a + 3b + 2a + b$

4  $4pq - pq + 3qp$

6  $2y + 5 - 2y$

## SKILL DRILL 2 Medium basics: keep practising

28 questions

## YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 28

Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–16: simplify. Collect the like terms.

1  $c - 2y - 7c$   
\_\_\_\_\_

2  $b - 7k - 3b$   
\_\_\_\_\_

3  $8a - 7b - a + 9$   
\_\_\_\_\_

4  $7c - 9p - c$   
\_\_\_\_\_

5  $7m + 4n + 3m + 5$   
\_\_\_\_\_

6  $5n - 9x + 7n$   
\_\_\_\_\_

7  $n - 8t + 2n$   
\_\_\_\_\_

8  $3a - 8k + 9a$   
\_\_\_\_\_

9  $9c - 8y - 5c$   
\_\_\_\_\_

10  $7k - p - 4k + 9$   
\_\_\_\_\_

11  $b + x + 5b + 6$   
\_\_\_\_\_

12  $c - 4y + 5c$   
\_\_\_\_\_

13  $3b - 8x + 5b$   
\_\_\_\_\_

14  $7a - t - 8a + 4$   
\_\_\_\_\_

15  $6m - 5n + 7m$   
\_\_\_\_\_

16  $m + 7p + 5m + 1$   
\_\_\_\_\_

## ROUND 2 Questions 17–28: simplify.

17  $9mp + 6m^2 - 4mp + 9m^2$   
\_\_\_\_\_

18  $8mp - 2m^2 - 9mp + 6m^2$   
\_\_\_\_\_

19  $7mt - 8m^2 - 7mt + m^2$   
\_\_\_\_\_

20  $8cp - 4c^2 - 4cp + 5c^2$   
\_\_\_\_\_

21  $2bt - 4b^2 - 7bt + 7b^2$   
\_\_\_\_\_

22  $7bt - b^2 - 2bt + 7b^2$   
\_\_\_\_\_

23  $7kt - 5k^2 - 5kt + 2k^2$   
\_\_\_\_\_

24  $9an + 3a^2 - 2an + 4a^2$   
\_\_\_\_\_

25  $4ax + 5a^2 - 3ax + 8a^2$   
\_\_\_\_\_

26  $6ky - 5k^2 - ky + 8k^2$   
\_\_\_\_\_

27  $9mp - 8m^2 - 6mp + 5m^2$   
\_\_\_\_\_

28  $4ay + 5a^2 - 6ay + 2a^2$   
\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Simplify. Collect each kind of term separately.

**a**  $5h^2 + 6h + 8h^2 + 11h$

Your teacher works this out. Copy every step.

**b**  $13h^3 - 6h - 8h - 11h^3$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Simplify.

4 QUESTIONS

**REMEMBER** Keep the sign in front of each term.

**1**  $5h^2 + 6h + 8h^2 + 11h$

**3**  $3ab - 2a + 5ab + 7a - b$

**2**  $13h^3 - 6h - 8h - 11h^3$

**4** A triangle has sides  $2x$ ,  $3x + 1$  and  $x + 4$ . Write and simplify its perimeter.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Simplify  $6k + 3k$ .

**Level 2**

Simplify  $4a + 7 - a + 2$ .

**Level 3**

Simplify  $3x^2 + 5x - x^2 - 8x$ .

**Extension · Stretch yourself****Worked example**Simplify  $4x^2y - 3xy^2 + x^2y + 5xy^2 - xy$ .

1. Like terms:  $4x^2y + x^2y$
2.  $-3xy^2 + 5xy^2$
3.  $-xy$  has no partner

Answer:  $5x^2y + 2xy^2 - xy$ **YOU DO**

On your own: read the worked example, then try these.

- E1**
- Simplify
- $7a^2 - 3a + 2 - a^2 + 5a - 8$
- .

- E2**
- Find the missing term:
- $5m + ? - 2m = 9m$

**Lesson summary**

Missed the lesson? Start here.

- Remember**
1. Like terms have exactly the same letters and powers.
  2. Add or subtract only like terms.
  3. Keep the sign in front of each term.
  4. Write the answer with the highest power first.

**Level 1**

$$3m + 4m$$

1.  $3 + 4 = 7$

Answer:  $7m$ **Level 2**

$$7y - 4y - y$$

1.  $7 - 4 - 1$

Answer:  $2y$ **Level 3**

$$6uv + 4 + 12uv$$

1.  $6 + 12 = 18$

Answer:  $18uv + 4$ **Extension**

$$5h^2 + 6h + 8h^2 + 11h$$

1.  $h^2$  terms, then  $h$  terms

Answer:  $13h^2 + 17h$ 

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LESSON  
5.08

## Multiplying terms

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- multiply algebraic terms

## Success criteria

I can:

- ☐ multiply the numbers first, then the letters
- ☐ write repeated letters as powers
- ☐ use the sign rules for negative terms

**Key terms** Write the definition as your teacher explains it.**Coefficient****Power**

Notes

## Level 1 · Getting started

Easy · 20 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Simplify.

**a**  $3m \times 4m$

Your teacher works this out. Copy every step.

**b**  $a \times a \times a \times a$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Simplify.

14 QUESTIONS

**REMEMBER** Multiply the numbers (coefficients).

**1**  $3 \times 4m$

\_\_\_\_\_

**2**  $2a \times 5$

\_\_\_\_\_

**3**  $a \times b$

\_\_\_\_\_

**4**  $a \times a \times a \times a$

\_\_\_\_\_

**5**  $x \times 7 \times y$

\_\_\_\_\_

**6**  $3m \times 4m$

\_\_\_\_\_

Questions 7–14: simplify.

**7**  $5 \times 2x$

\_\_\_\_\_

**8**  $3a \times 4$

\_\_\_\_\_

**9**  $x \times y \times 2$

\_\_\_\_\_

**10**  $m \times m$

\_\_\_\_\_

**11**  $2p \times 3q$

\_\_\_\_\_

**12**  $4k \times k$

\_\_\_\_\_

**13**  $-3 \times 5y$

\_\_\_\_\_

**14**  $-2a \times (-4b)$

\_\_\_\_\_

## SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

## YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 40

Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–24: simplify. Multiply the numbers, then the letters.

1  $7k \times 7p$

\_\_\_\_\_

2  $5 \times 9t$

\_\_\_\_\_

3  $2b \times 8c$

\_\_\_\_\_

4  $3b \times 8y$

\_\_\_\_\_

5  $4 \times 5c$

\_\_\_\_\_

6  $7b \times 8n$

\_\_\_\_\_

7  $3c \times 2p$

\_\_\_\_\_

8  $8c \times 4p$

\_\_\_\_\_

9  $5 \times 3a$

\_\_\_\_\_

10  $4 \times 4m$

\_\_\_\_\_

11  $9 \times 4k$

\_\_\_\_\_

12  $5a \times 3c$

\_\_\_\_\_

13  $6 \times 4k$

\_\_\_\_\_

14  $3 \times 6p$

\_\_\_\_\_

15  $8a \times 7b$

\_\_\_\_\_

16  $4a \times 8c$

\_\_\_\_\_

17  $7a \times 7p$

\_\_\_\_\_

18  $6k \times 4p$

\_\_\_\_\_

19  $9k \times 3y$

\_\_\_\_\_

20  $3 \times 4k$

\_\_\_\_\_

21  $4 \times 5b$

\_\_\_\_\_

22  $2k \times 8y$

\_\_\_\_\_

23  $3c \times 6t$

\_\_\_\_\_

24  $9p \times 5t$

\_\_\_\_\_

## ROUND 2 Questions 25–40: simplify. Use index notation.

25  $ct \times t$

\_\_\_\_\_

26  $cm \times m$

\_\_\_\_\_

27  $m \times m$

\_\_\_\_\_

28  $n \times t \times n$

\_\_\_\_\_

29  $p \times p$

\_\_\_\_\_

30  $cx \times x$

\_\_\_\_\_

31  $k \times k$

\_\_\_\_\_

32  $c \times c \times c$

\_\_\_\_\_

33  $k \times k \times k$

\_\_\_\_\_

34  $a \times a$

\_\_\_\_\_

35  $ck \times k$

\_\_\_\_\_

36  $b \times p \times b$

\_\_\_\_\_

37  $cy \times y$

\_\_\_\_\_

38  $a \times a \times a$

\_\_\_\_\_

39  $b \times x \times b$

\_\_\_\_\_

40  $b \times b$

\_\_\_\_\_



**YOU DO**

On your own: **Easy** Set B. Simplify.

6 QUESTIONS

1  $6p \times 2q$

4  $5ab \times 2a$

2  $9v \times 4 \times 5v$

5  $4x \times 3x \times x$

3  $k \times k \times 2$

6  $2c \times 2c \times 2c$



**Level 2 · Building up**

Medium · 6 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

**Simplify.**

a  $9v \times 4 \times 5v$

Your teacher works this out. Copy every step.

b  $3pq \times 5pq$

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** Set C. Simplify. Watch the signs.

6 QUESTIONS

**REMEMBER** Multiply the letters, in alphabetical order.

1  $-7d \times (-3d)$

3  $3x \times (-x)$

5  $-3x^2y \times 2x$

2  $-2a \times 6b$

4  $-4m \times (-m) \times 2$

6  $5 \times (-t) \times (-t)$

## SKILL DRILL 2 Medium basics: keep practising

32 questions

## YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 32 Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–16: simplify.

1  $3k \times 6km$   
\_\_\_\_\_

2  $4mp \times 5mp$   
\_\_\_\_\_

3  $6km \times 3km$   
\_\_\_\_\_

4  $4c \times 5c$   
\_\_\_\_\_

5  $3c \times 6c$   
\_\_\_\_\_

6  $5k \times 2a$   
\_\_\_\_\_

7  $4ak \times 5ak$   
\_\_\_\_\_

8  $5y \times 6n$   
\_\_\_\_\_

9  $6m \times 5y$   
\_\_\_\_\_

10  $2py \times 3py$   
\_\_\_\_\_

11  $2nx \times 2x$   
\_\_\_\_\_

12  $5k \times 2k$   
\_\_\_\_\_

13  $5n \times 2b$   
\_\_\_\_\_

14  $3c \times 6ck$   
\_\_\_\_\_

15  $3p \times 4y$   
\_\_\_\_\_

16  $4y \times 3c$   
\_\_\_\_\_

## ROUND 2 Questions 17–32: simplify. Take care with the signs.

17  $-6c \times (-3n)$   
\_\_\_\_\_

18  $-4t \times 6y$   
\_\_\_\_\_

19  $-3a \times (-5n)$   
\_\_\_\_\_

20  $6a \times (-5k)$   
\_\_\_\_\_

21  $-6k \times 3m$   
\_\_\_\_\_

22  $5b \times 3n$   
\_\_\_\_\_

23  $-6m \times (-2n)$   
\_\_\_\_\_

24  $x \times (-4y)$   
\_\_\_\_\_

25  $6c \times (-6m)$   
\_\_\_\_\_

26  $5k \times (-6y)$   
\_\_\_\_\_

27  $2b \times 3x$   
\_\_\_\_\_

28  $t \times 6x$   
\_\_\_\_\_

29  $-n \times 4p$   
\_\_\_\_\_

30  $b \times 2t$   
\_\_\_\_\_

31  $-m \times 4x$   
\_\_\_\_\_

32  $-6m \times 3x$   
\_\_\_\_\_

**Level 3 · Challenge**

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Simplify. Watch the signs.

**a**  $-7d \times (-3d)$

Your teacher works this out. Copy every step.

**b**  $lmn \times lmn \times lmn$

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Simplify.

4 QUESTIONS

**REMEMBER** Same letter repeated → use a power.

**1**  $3pq \times 5pq$

**3**  $2a^2 \times 3a^3$

**2**  $lmn \times lmn \times lmn$

**4** A rectangle is  $3x$  cm long and  $2y$  cm wide. Write its area.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

**YOU DO** **Show off your skill**

Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

**Level 1**

Simplify  $5 \times 3k$ .

**Level 2**

Simplify  $4a \times 6ab$ .

**Level 3**

Simplify  $-2x \times 3x \times (-y)$ .

## Extension · Stretch yourself

## Worked example

Simplify  $(2ab)^3$ .

- $2ab \times 2ab \times 2ab$
- Numbers:  $2 \times 2 \times 2 = 8$
- $a^3b^3$

Answer:  $8a^3b^3$ 

## YOU DO

On your own: read the worked example, then try these.

- E1 Simplify
- $(-3m)^2$
- .

- E2 Find the missing term:
- $4x \times ? = 20x^2y$
- .

## Lesson summary

Missed the lesson? Start here.

- Remember**
1. Multiply the numbers (coefficients).
  3. Same letter repeated  $\rightarrow$  use a power.

2. Multiply the letters, in alphabetical order.
4. Signs: same  $\rightarrow$  +, different  $\rightarrow$  -.

## Level 1

$$3m \times 4m$$

- 1.
- $3 \times 4 = 12$
- ,
- $m \times m = m^2$

Answer:  $12m^2$ 

## Level 2

$$9v \times 4 \times 5v$$

- 1.
- $9 \times 4 \times 5 = 180$

Answer:  $180v^2$ 

## Level 3

$$-7d \times (-3d)$$

- 1.
- $- \times - = +$

Answer:  $21d^2$ 

## Extension

$$(2ab)^3$$

- 1.
- $8, a^3, b^3$

Answer:  $8a^3b^3$ 

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LESSON  
5.09

## Dividing terms

Date \_\_\_\_\_

## Learning intentions

We are learning to:

- divide algebraic terms

## Success criteria

I can:

- ☐ write a division as a fraction
- ☐ cancel common factors in the numbers and the letters
- ☐ simplify answers that are still fractions

**Key terms** Write the definition as your teacher explains it.**Quotient****Cancel**

Notes

## Level 1 · Getting started

Easy · 18 questions

**WE DO** With your teacher: copy what they write on the board.**Example 1** Simplify.

**a**  $15m \div 3$

**b**  $9v \div v$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Easy** Set A. Questions 1–6: Simplify.

12 QUESTIONS

**REMEMBER** Write the division as a fraction.

**1**  $15m \div 3$

**2**  $9v \div v$

**3**  $\frac{49y}{7}$

**4**  $8ab \div 2$

**5**  $20k \div 5k$

**6**  $\frac{6pq}{q}$

Questions 7–12: simplify.

**7**  $12a \div 4$

**8**  $18m \div 6$

**9**  $10x \div x$

**10**  $24pq \div 8p$

**11**  $6ab \div 3b$

**12**  $15k \div 5k$

## SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

## YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 40

Mark it with your teacher. Aim to beat your time next lesson.

## ROUND 1 Questions 1–24: simplify.

1  $72p \div 8$

\_\_\_\_\_

2  $8a \div 4$

\_\_\_\_\_

3  $24t \div 6$

\_\_\_\_\_

4  $54n \div 9$

\_\_\_\_\_

5  $18t \div 6$

\_\_\_\_\_

6  $35m \div 7$

\_\_\_\_\_

7  $32k \div 8$

\_\_\_\_\_

8  $18b \div 3$

\_\_\_\_\_

9  $18x \div 9$

\_\_\_\_\_

10  $49n \div 7$

\_\_\_\_\_

11  $49x \div 7$

\_\_\_\_\_

12  $25m \div 5$

\_\_\_\_\_

13  $6p \div 2$

\_\_\_\_\_

14  $48c \div 8$

\_\_\_\_\_

15  $56c \div 7$

\_\_\_\_\_

16  $9y \div 3$

\_\_\_\_\_

17  $18t \div 9$

\_\_\_\_\_

18  $15k \div 5$

\_\_\_\_\_

19  $21p \div 3$

\_\_\_\_\_

20  $54c \div 9$

\_\_\_\_\_

21  $30a \div 5$

\_\_\_\_\_

22  $36k \div 4$

\_\_\_\_\_

23  $36b \div 4$

\_\_\_\_\_

24  $18m \div 9$

\_\_\_\_\_

## ROUND 2 Questions 25–40: simplify.

25  $c^2 \div c$

\_\_\_\_\_

26  $ty \div t$

\_\_\_\_\_

27  $cy \div c$

\_\_\_\_\_

28  $am \div m$

\_\_\_\_\_

29  $bn \div n$

\_\_\_\_\_

30  $cy \div y$

\_\_\_\_\_

31  $n^2 \div n$

\_\_\_\_\_

32  $xy \div y$

\_\_\_\_\_

33  $ct \div t$

\_\_\_\_\_

34  $bt \div b$

\_\_\_\_\_

35  $am \div a$

\_\_\_\_\_

36  $bk \div k$

\_\_\_\_\_

37  $ty \div y$

\_\_\_\_\_

38  $bx \div b$

\_\_\_\_\_

39  $cn \div n$

\_\_\_\_\_

40  $bn \div b$

\_\_\_\_\_

**YOU DO**

On your own: **Easy** Set B. Simplify.

6 QUESTIONS

1  $\frac{12x^2y}{4xy}$

4  $24t^3 \div 8t$

2  $18m^2 \div 6m$

5  $\frac{3x}{12}$

3  $\frac{10ab}{5b}$

6  $\frac{4a}{6ab}$



**Level 2 · Building up**

Medium · 6 questions

**WE DO**

With your teacher: copy what they write on the board.

**Example 2**

Simplify. Write the division as a fraction.

a  $\frac{49y}{7}$

Your teacher works this out. Copy every step.

b  $\frac{12x^2y}{4xy}$

Your teacher works this out. Copy every step.

**YOU DO**

On your own: **Medium** Set C. Simplify.

6 QUESTIONS

**REMEMBER** Cancel the numbers by their HCF.

1  $32de^2 \div 10de$

3  $\frac{-15x^2}{-5x}$

5  $\frac{14pq^2}{21pq}$

2  $\frac{-20mn}{4m}$

4  $\frac{9a^2b^2}{3ab}$

6  $-36y^3 \div 4y^2$

SKILL DRILL 2 Medium basics: keep practising

32 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time \_\_\_\_\_ Finish time \_\_\_\_\_ Score \_\_\_\_\_ / 32

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: simplify.

1  $35p^2 \div 5p$

\_\_\_\_\_

2  $8kx \div 4x$

\_\_\_\_\_

3  $10bp \div 2b$

\_\_\_\_\_

4  $12ct \div 4c$

\_\_\_\_\_

5  $18ab \div 2b$

\_\_\_\_\_

6  $54ab^2 \div 6b$

\_\_\_\_\_

7  $18pt \div 2p$

\_\_\_\_\_

8  $15b^2 \div 3b$

\_\_\_\_\_

9  $6m^2 \div 3m$

\_\_\_\_\_

10  $18ck \div 2c$

\_\_\_\_\_

11  $24mt \div 3t$

\_\_\_\_\_

12  $24a^2 \div 3a$

\_\_\_\_\_

13  $36a^2b \div 4a$

\_\_\_\_\_

14  $35mt \div 5m$

\_\_\_\_\_

15  $12bn \div 4b$

\_\_\_\_\_

16  $8cm^2 \div 2m$

\_\_\_\_\_

ROUND 2 Questions 17–32: simplify the fraction.

17  $\frac{24p^2}{6p}$

\_\_\_\_\_

18  $\frac{6n}{2n}$

\_\_\_\_\_

19  $\frac{27n}{3n}$

\_\_\_\_\_

20  $\frac{42t^2}{6t}$

\_\_\_\_\_

21  $\frac{35mt}{5m}$

\_\_\_\_\_

22  $\frac{40x}{5x}$

\_\_\_\_\_

23  $\frac{4y^2}{2y}$

\_\_\_\_\_

24  $\frac{18nx}{3x}$

\_\_\_\_\_

25  $\frac{28t}{4t}$

\_\_\_\_\_

26  $\frac{54ny}{6y}$

\_\_\_\_\_

27  $\frac{12mx}{3m}$

\_\_\_\_\_

28  $\frac{24px}{3x}$

\_\_\_\_\_

29  $\frac{20m}{4m}$

\_\_\_\_\_

30  $\frac{15a}{5a}$

\_\_\_\_\_

31  $\frac{40k}{5k}$

\_\_\_\_\_

32  $\frac{30n}{6n}$

\_\_\_\_\_



Level 3 · Challenge

Challenging · 4 questions · then try the extension

**WE DO** With your teacher: copy what they write on the board.

**Example 3** Simplify.

a  $32de^2 \div 10de$

b  $\frac{-3 \times 2 y \times 2x}{3x \times (-x)}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

**YOU DO** On your own: **Challenging** Set D. Simplify.

4 QUESTIONS

**REMEMBER** Cancel letters that are on top and bottom.

1  $\frac{-3 \times 2 y \times 2x}{3x \times (-x)}$

3  $\frac{10m2n}{2m \times 5n}$

2  $\frac{4a \times 6b}{8ab}$

4 A rectangle has area  $24xy \text{ cm}^2$  and length  $6x \text{ cm}$ . Find its width.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

**YOU DO** **Show off your skill** Choose **ONE** question. Show your working.

Name \_\_\_\_\_ Class \_\_\_\_\_

Level 1

Simplify  $21p \div 7$ .

Level 2

Simplify  $\frac{16ab^2}{4b}$ .

Level 3

Simplify  $\frac{-10 \times 3}{15x}$ .

**Extension · Stretch yourself****Worked example**Simplify  $\frac{6a3b \times 4ab2}{12a2b2}$ .

1. Top:  $24a^4b^3$

2.  $\div 12a^2b^2$

Answer:  $2a^2b$ **YOU DO**

On your own: read the worked example, then try these.

E1 Simplify  $\frac{(3m)2}{6m}$ .

E2 Simplify  $\frac{8 \times 2y \times 3y}{6xy \times 2x}$ .

**Lesson summary**

Missed the lesson? Start here.

**Remember** 1. Write the division as a fraction.  
3. Cancel letters that are on top and bottom.2. Cancel the numbers by their HCF.  
4. Signs: same  $\rightarrow +$ , different  $\rightarrow -$ .**Level 1**

$15m \div 3$

1.  $15 \div 3 = 5$

Answer:  $5m$ **Level 2**

$\frac{12 \times 2y}{4xy}$

1.  $12 \div 4 = 3$ , cancel x and y

Answer:  $3x$ **Level 3**

$32de^2 \div 10de$

1.  $32/10 = 16/5$ , cancel d and e

Answer:  $\frac{16e}{5}$ **Extension**

$\frac{6a3b \times 4ab2}{12a2b2}$

1.  $24 \div 12 = 2$

Answer:  $2a^2b$ 

This space is left blank on purpose. It is the back of your exit ticket.